## 5.3 Structure of Glucose, Fructose and Galactose

### 5.3.1 Glucose:

Glucose has the molecular formula ($C_6H_{12}O_6$) and has a six member ring. Glucose may be represented by the following open chain structure. But in solution it exists only as a six membered ring structure called pyranose form. Glucose is known as grape sugar.

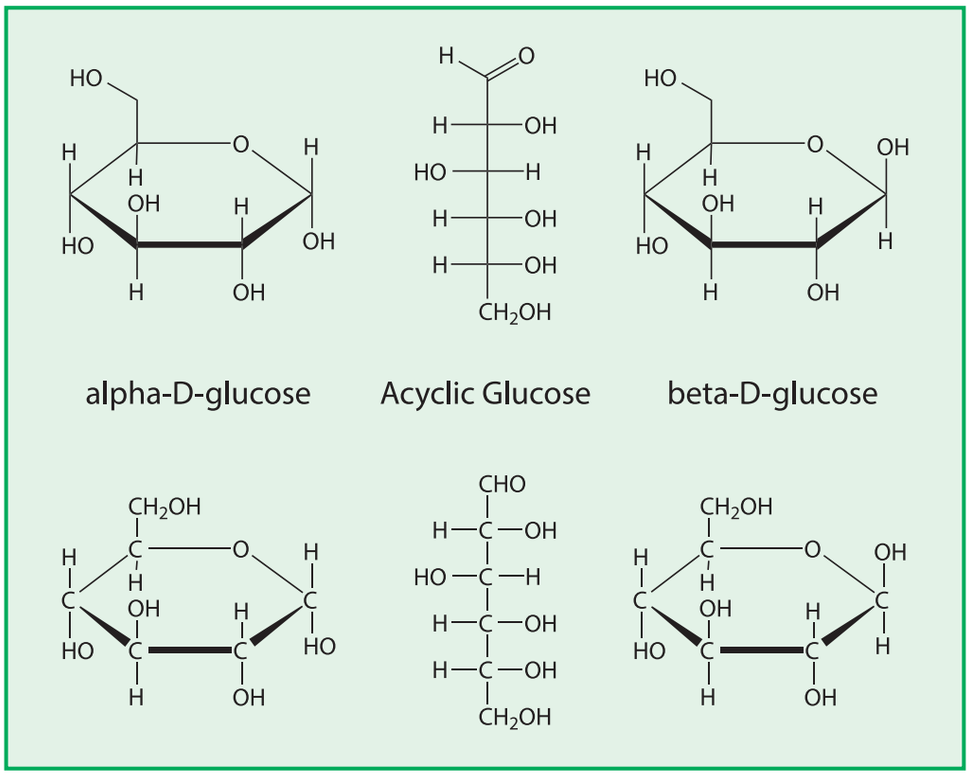



*Figure 5.3 Structure of Glucose*




Glucose is eaten, absorbed into the blood stream, and makes it way to the liver where it is broken down to supply energy to the entire body. This breaking down process requires insulin.

### 5.3.2 Fructose

Fructose has the molecular formula ($C_6H_{12}O_6$) and has a five member ring. Fructose exists mostly as a five membered ring structure called "furanose form". Fructose is known as the fruit sugar as its make source in the diet is fruits and vegetables. Honey is also a good source.



Unit 5.indd 126 21-05-2018 15:10:27